

Mark Mazumder

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Research Objective

My research establishes data-centric learning and evaluation methods, targeting problems with limited training data. My current focus is on imbuing domain expertise in language models on topics outside of the pretraining data distribution. My research purview spans multiple modalities (text, code, vision, and speech), with an emphasis on improving feature representations.

Education

- o **Harvard University**, Cambridge, MA. *PhD Candidate in Electrical Engineering*. Fall 2020-Present
Advisor: Professor Vijay Janapa Reddi, Edge Computing Lab [\[link\]](#) GPA: 4.0
- o **Harvard University**, Cambridge, MA. *Bachelor of Arts in Computer Science*. 2009

Selected Publications

- o *QuArch: A Benchmark for Evaluating LLM Reasoning in Computer Architecture*. S. Prakash, A. Cheng, **M. Mazumder**, A. Tschand, V. Gohil, J. J. Ma, J. Yik, Z. Wan, J. Quaye, E. L. Alvanaki, A. Kumar, C. Mazumdar, T. Khare, A. Ingare, I. Uchendu, R. Ghosal, A. Tyagi, C. Wang, A. M. Garavagno, S. Y. Gu, A. Guo, G. Hur, L. Carloni, T. Krishna, A. Nayak, A. Yazdanbakhsh, V. Janapa Reddi. ICML 2026. [\[OpenReview\]](#)
- o *Wake Vision: A Tailored Dataset and Benchmark Suite for TinyML Computer Vision Applications*. C. Banbury, E. Njor, A. Garavagno, **M. Mazumder**, P. Warden, M. Kudlur, N. Jeffries, X. Fafoutis, V. Janapa Reddi. Journal of Data-centric Machine Learning Research, 2026. [\[DMLR Proceedings\]](#)
- o *DataPerf: Benchmarks for Data-Centric AI Development*. **M. Mazumder**, C. Banbury, X. Yao, B. Karlaš, W. Rojas, S. Diamos, G. Diamos, L. He, D. Kiela, D. Jurado, D. Kanter, R. Mosquera, J. Ciro, L. Aroyo, B. Acun, S. Eyuboglu, A. Ghorbani, E. Goodman, T. Kane, C. Kirkpatrick, T. Kuo, J. Mueller, T. Thrush, J. Vanschoren, M. Warren, A. Williams, S. Yeung, N. Ardalani, P. Paritosh, C. Zhang, J. Zou, C. Wu, C. Coleman, A. Ng, P. Mattson, V. Janapa Reddi. NeurIPS 2023 Track on Datasets and Benchmarks. [\[NeurIPS Proceedings\]](#)
- o *Multilingual Spoken Words Corpus*. **M. Mazumder**, S. Chitlangia, C. Banbury, Y. Kang, J. Ciro, K. Achorn, D. Galvez, M. Sabini, P. Mattson, D. Kanter, G. Diamos, P. Warden, J. Meyer, V. Janapa Reddi. NeurIPS 2021 Track on Datasets and Benchmarks. [\[NeurIPS Proceedings\]](#) [\[Harvard News\]](#) [\[Dataset Download\]](#)
- o *Few-shot Keyword Spotting in Any Language*. **M. Mazumder**, C. Banbury, J. Meyer, P. Warden, V. Janapa Reddi. INTERSPEECH 2021. [\[arXiv:2104.01454\]](#)
- o *Towards an Autonomous Aerial Survey and Planning System for Humanitarian Aid and Disaster Response*. R. Allen, **M. Mazumder**. IEEE Aerospace Conference 2020. [\[doi:10.1109/AERO47225.2020.9172766\]](#)
- o *Active Rendezvous for Multi-Robot Pose Graph Optimization using Sensing over Wi-Fi*. W. Wang, N. Jadhav, P. Vohs, N. Hughes, **M. Mazumder**, S. Gil. International Symposium on Robotics Research (ISRR) 2019. [\[arXiv:1907.05538\]](#)
- o *Guaranteeing Spoof-Resilient Multi-Robot Networks*. S. Gil, S. Kumar, **M. Mazumder**, D. Katabi, D. Rus. Autonomous Robots 2017. [\[Journal Article\]](#) [\[MIT News\]](#) (subsumes our [RSS 2015 paper](#))

Work Experience

Apple, Seattle WA, Cambridge MA

Summer 2022, Fall 2022, Summer 2023

AI/ML Intern, *Machine Intelligence: Sensing*

Researching novel learning methods for improving supervision on multimodal time-series data.

- o Developed interpretability algorithms for multimodal fusion transformers for video classification tasks
- o Characterized the effects of feature noise, label noise, and fusion strategies on multimodal data
- o Designed curriculum learning algorithms to improve model representations for higher-noise modalities

Landing AI, Palo Alto CA

Summer 2021

Intern, *Continuous Learning Team*

Developed and prototyped novel anomaly detection methods for visual inspection.

- o Collaborated cross-functionally to distill high-level customer needs into concrete engineering requirements
- o Leveraged data-centric ML and error analysis to systematically innovate and improve anomaly detection model
- o Kick-started integration and deployment of anomaly detection models into customer-facing product features

MIT Lincoln Laboratory, Lexington MA

2012-2020

Associate Staff, *Group 104: Artificial Intelligence Software Architecture and Algorithms*.

- o Served as co-PI on two autonomous navigation research efforts:
 - *Transferring Multi-Robot Learning through Virtual and Augmented Reality for Rapid Disaster Response*. Deploys Sim2Real visual navigation reinforcement learning policies without domain randomization.
 - *Resilient Perception in Degraded Environments*. Multi-agent mapping utilizing outlier-robust pose graph optimization and the physics of wireless signals for efficient coordination.
- o Engineering roles include SmallSat flight software development, test and evaluation engineering for a DARPA program, and high-scale network traffic generation.

Workshop Activities

- o *Co-organizer*: DataPerf Workshop. ICML 2022. [\[link\]](#).
- o *Co-organizer*: Data-Centric AI Workshop. NeurIPS 2021. [\[link\]](#).
- o *Invited Talk* 1000 Words in 1000 Languages. MLPerf-Bench, HPCA 2021 [\[link\]](#).
- o *Co-organizer*: Perception, Action, Learning: From Metric-Semantic Scene Understanding to High-Level Task Execution. IEEE International Conference on Robotics and Automation (ICRA) 2020. [\[link\]](#)
- o *Tutorial* Safe Client/Server Web Development with Haskell. Mark Mazumder, Tim Braje. IEEE SecDev 2016.

Teaching Activities

- o *Head TF*, **Harvard University CS249r: Tiny Machine Learning**, Prof. Vijay Janapa Reddi Fall 2022
- o *Contributor*, **Applications of TinyML**, EdX [\[edx.org\]](#), Prof. Vijay Janapa Reddi Fall 2020
- o *Instructor*, **MIT 6.A01 Autonomous Racecar Robotics Seminar**, Prof. Sertac Karaman Fall 2019
Website: https://markmaz.com/racecar_fall19/
- o *Instructor*, **MIT Beaver Works Summer Institute: Autonomous Air Vehicle Racing** 2018-2019
Website: <https://bws-i-uav.github.io/website/index.html>
- o *Instructor*, **MIT NEET-AM Machine Learning Labs**, Prof. Sertac Karaman Spring 2019
News: <https://blogs.nvidia.com/blog/2019/11/21/mit-quadro-data-science-workstations/>
- o *Instructor*, **MIT 6.S184/16.S685 RACECAR**, Independent Activities Period January 2019
- o *Instructor*, **MIT 16.S688 Autonomous Machines Seminar**, Prof. Sertac Karaman Fall 2018
- o *Lab Assistant*, **MIT 16.30/16.31 Feedback Control Systems**, Prof. Sertac Karaman Fall 2018
- o *Instructor*, **MIT 6.A01 Autonomous Racecar Robotics Seminar**, Prof. Sertac Karaman Fall 2017
- o *Lab Assistant*, **MIT 6.829 Computer Networks**, Prof. Dina Katabi Fall 2015
- o *Teaching Fellow*, **Harvard University CS161: Operating Systems**, Prof. Margo Seltzer Spring 2011